

V4.2 Toy Model 3C

Recovering the Kinetic-Term Structure: Toward Spatial Wave Dynamics

Rev A - Exploratory Mathematical Compatibility Report

Date: 2026.05.25 | Continuation after Toy Model 3B

Status: Toy Model 3C tests whether spatial phase twisting produces the structural form of the quantum momentum and kinetic-energy operators. It does not derive mass, Planck's constant, or the physical Hamiltonian from first principles.

1. Purpose

Toy Model 3B showed that observable time evolution can be represented as hidden transverse phase rotation. Toy Model 3C now adds space. If phase changes along a spatial coordinate x , the state becomes a spatial wave. The test is whether spatial derivatives of this phase-twisting state reproduce the structural form of quantum momentum and kinetic energy.

The target is structural compatibility with the free-particle Schrodinger equation. This is not yet a full derivation of physical mass, Planck's constant, or a real particle Hamiltonian.

2. Spatial phase twisting

Start with the real transverse phase-plane rotation operator from 3A and 3B:

$$J = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$J^2 = -I$$

Now allow the phase angle to vary with spatial position:

$$\phi(x, t) = kx - \omega t$$

Term reminders:

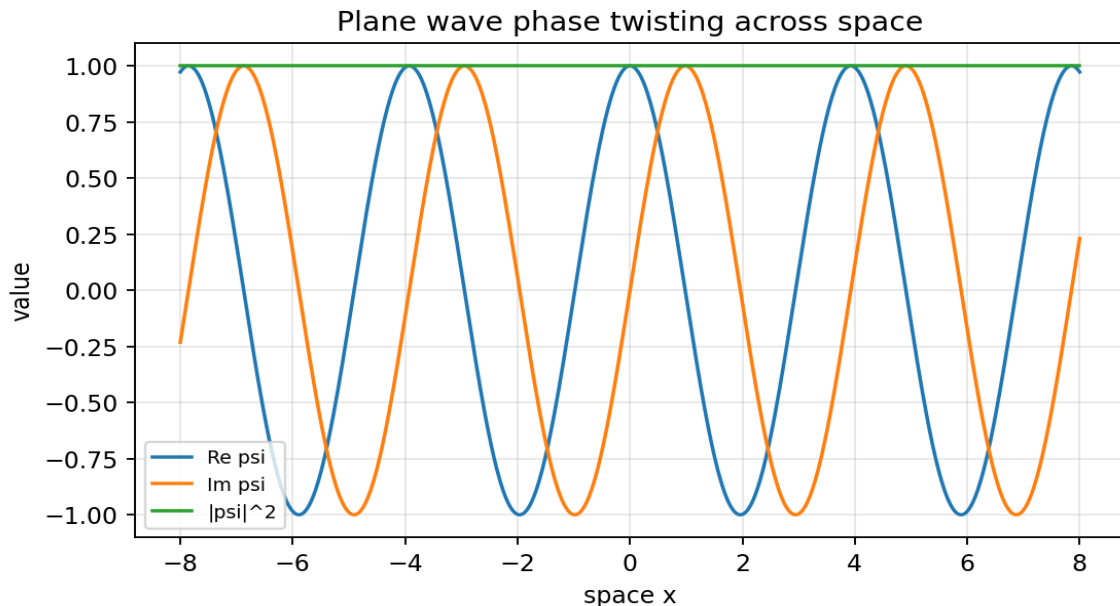
- x is a one-dimensional spatial coordinate.
- k is wave number, meaning spatial phase-twist rate in radians per unit length.
- ω is angular frequency, meaning temporal phase-rotation rate.
- ϕ is the phase angle.
- J is the real 90-degree rotation operator in the hidden transverse phase plane.

The real phase-plane state is:

$$v(x, t) = R(kx - \omega t) v(0)$$

In complex shorthand this is the plane wave:

$$\psi(x, t) = \exp(i(kx - \omega t))$$



3. First spatial derivative: momentum structure

Taking the first spatial derivative gives, in the real phase-plane representation:

$$d v / dx = k J v$$

In complex shorthand:

$$d \psi / dx = i k \psi$$

The standard quantum momentum operator in one dimension is:

$$\hat{p} = -i \hbar d/dx$$

Apply it to the plane wave:

$$\begin{aligned} \hat{p} \psi &= -i \hbar (i k \psi) \\ &= \hbar k \psi \end{aligned}$$

So the plane wave is an eigenstate of the momentum operator with momentum:

$$p = \hbar k$$

Plain meaning: spatial phase-twist rate k maps to momentum. Higher spatial twist means higher momentum.

4. Second spatial derivative: kinetic structure

Taking the second spatial derivative applies J twice:

$$\begin{aligned} d^2 v / dx^2 &= k J (d v / dx) \\ &= k J (k J v) \\ &= k^2 J^2 v \\ &= -k^2 v \end{aligned}$$

In complex shorthand:

$$d^2 \psi / dx^2 = -k^2 \psi$$

This negative scaling is the key kinetic-term structure. The standard free-particle kinetic operator is:

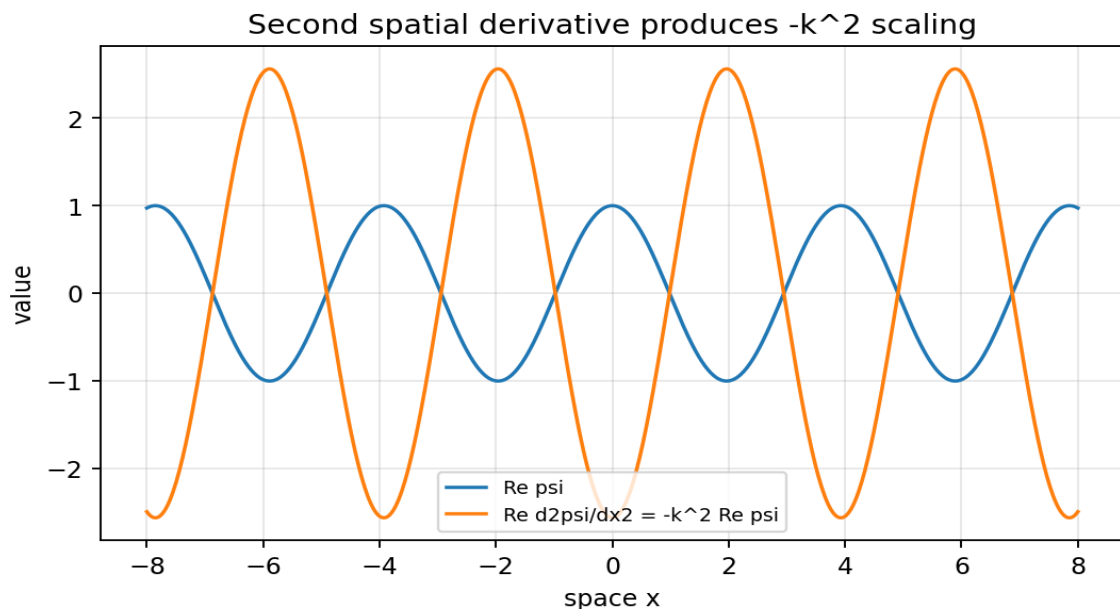
$$\hat{T} = -(\hbar^2 / 2m) d^2/dx^2$$

Apply it to the plane wave:

$$\begin{aligned} \hat{T} \psi &= -(\hbar^2 / 2m) (-k^2 \psi) \\ &= (\hbar^2 k^2 / 2m) \psi \end{aligned}$$

So the kinetic energy is:

$$E = \hbar^2 k^2 / 2m$$



5. Free-particle Schrodinger compatibility

From 3B, the temporal phase part gives:

$$i \hbar d \psi / dt = \hbar \omega \psi$$

The kinetic operator gives:

$$-(\hbar^2 / 2m) d^2 \psi / dx^2 = (\hbar^2 k^2 / 2m) \psi$$

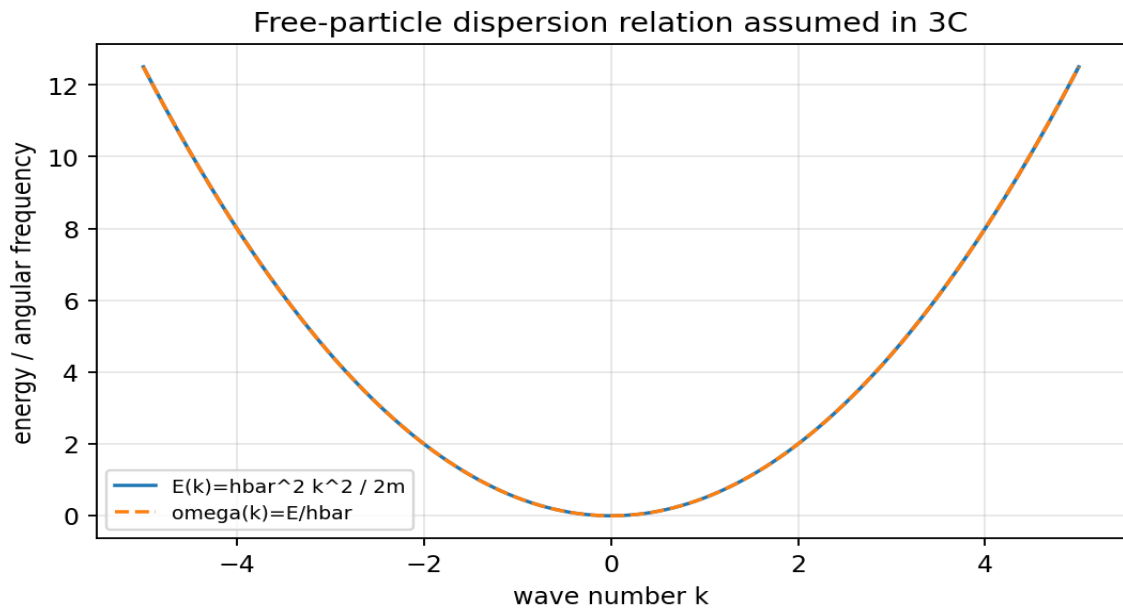
For the free particle, require the dispersion relation:

$$\hbar \omega = \hbar^2 k^2 / 2m$$

Then both sides match and the free-particle Schrodinger equation is recovered structurally:

$$i \hbar \frac{d \psi}{dt} = -(\hbar^2 / 2m) \frac{d^2 \psi}{dx^2}$$

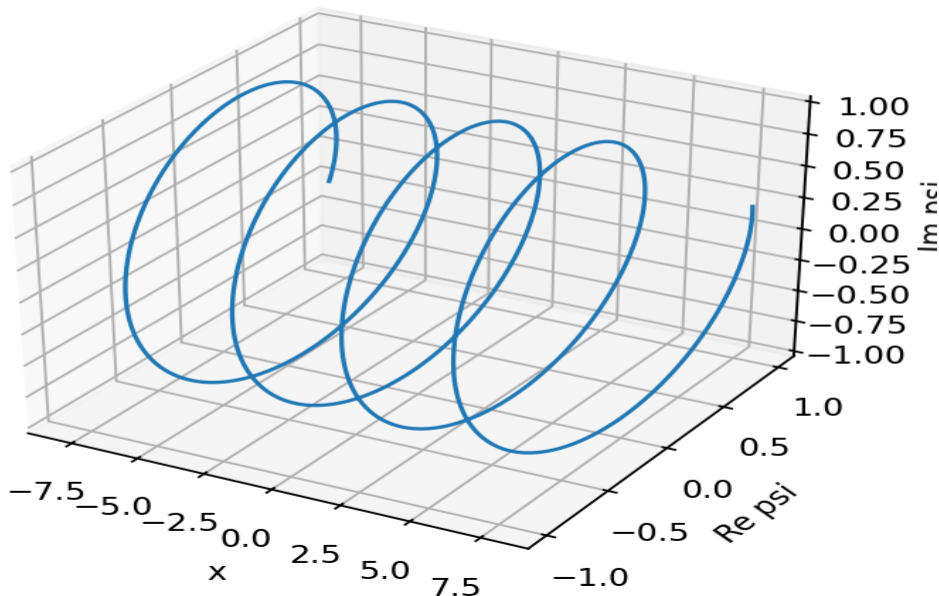
Guardrail: the dispersion relation uses the standard nonrelativistic kinetic-energy relation $E=p^2/2m$ together with $p=\hbar k$ and $E=\hbar \omega$. 3C does not derive mass m or $\hbar$ from first principles.



6. Geometric interpretation

The plane wave can be visualized as a phase helix across space. Moving along x rotates the hidden transverse phase vector. The stronger the twist per length, the larger k becomes.

Spatial phase helix: phase angle $\phi(x)=kx$



In 6D language, momentum is not introduced as a separate mystery. It is represented as the spatial rate of hidden phase rotation across the observer-accessible 3S direction.

7. Numerical checks

Check	Result
Random samples	10000
Max $J^2 + I$ error	0.000e+00

Check	Result
Max real first-derivative error	3.922e-09
Max real second-derivative error	1.421e-04
Max momentum-operator error	4.030e-09
Max kinetic-operator error	4.217e-05
Max Schrodinger residual	4.217e-05
Max dispersion relation error	0.000e+00
Max real norm error	2.220e-16

8. Interpretation inside the 6D Projection Framework

Toy Model 3C supports the following limited interpretation:

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spatial movement in 3S
-> hidden transverse phase twists along x
-> first derivative measures phase-gradient rate k
-> momentum structure  $p = \hbar k$ 
-> second derivative applies  $J$  twice
->  $J^2 = -I$  gives  $-k^2$  scaling
-> kinetic operator structure emerges

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This completes the first structural bridge from hidden phase geometry to the free-particle Schrodinger form.

9. What Toy Model 3C achieves

- It introduces spatial dependence into the hidden phase-plane model.
- It shows that a spatial phase gradient produces the momentum-operator structure.
- It shows that the second spatial derivative produces $-k^2$ through $J^2 = -I$.
- It recovers the structural form of the free-particle kinetic operator.
- It shows compatibility with the free-particle Schrodinger equation when the standard nonrelativistic dispersion relation is imposed.
- It interprets momentum as spatial rate of hidden phase twisting.

10. What Toy Model 3C does not achieve

- It does not derive physical mass m .
- It does not derive Planck's constant $\hbar$.
- It does not derive the nonrelativistic dispersion relation from deeper geometry.
- It does not derive potentials $V(x)$.
- It does not derive relativistic quantum mechanics.
- It does not derive quantum field theory or the Standard Model.
- It does not prove physical 3T existence.

11. Close-out statement for 3C

Toy Model 3C successfully constructs a structural bridge from spatial hidden phase twisting to the quantum momentum and kinetic-energy operators. A plane-wave phase field $\psi(x,t)=\exp(i(kx-\omega t))$ corresponds to real rotation in the hidden transverse phase plane. The first spatial derivative yields the momentum structure $\hat{p}=-i\hbar d/dx$ and $p=\hbar k$. The second derivative applies J twice, giving $J^2=-I$ and therefore $d^2\psi/dx^2=-k^2\psi$. With the standard free-particle dispersion relation $\hbar\omega=\hbar^2 k^2/(2m)$, the free-particle Schrodinger equation is recovered structurally. This is not a derivation of mass, $\hbar$, or the full Hamiltonian.

12. Recommended next chapter

The next logical chapter is Toy Model 3D: Adding Potentials as Position-Dependent Phase Rotation Rate.

1. Introduce $V(x)$ as a position-dependent contribution to local phase rotation rate.
2. Test whether $i\hbar d\psi/dt = [-(\hbar^2/2m)d^2/dx^2 + V(x)]\psi$ is recovered structurally.
3. Clarify whether $V(x)$ is an external input or can be interpreted as relational coupling density.

4. Check norm preservation for Hermitian potential terms.
5. Keep guardrails: do not claim derivation of force fields until $V(x)$ is linked to physical sources.

13. Rebind prompt for continuation

We are resuming after Toy Model 3C.

3C tested spatial wave dynamics and kinetic-term structure.

Phase field:

$$\phi(x,t) = kx - \omega t$$

Real hidden phase-plane state:

$$v(x,t) = R(kx - \omega t) v(0)$$

Complex shorthand:

$$\psi(x,t) = \exp(i(kx - \omega t))$$

First spatial derivative:

$$d\psi/dx = i k \psi$$

Momentum operator:

$$\hat{p} = -i \hbar d/dx$$

$$\hat{p} \psi = \hbar k \psi$$

Second spatial derivative:

$$d^2\psi/dx^2 = -k^2 \psi$$

Kinetic operator:

$$\hat{T} = -(\hbar^2/2m) d^2/dx^2$$

$$\hat{T} \psi = (\hbar^2 k^2/2m) \psi$$

With standard dispersion:

$$\hbar \omega = \hbar^2 k^2 / 2m$$

Free-particle Schrodinger structure:

$$i \hbar d\psi/dt = -(\hbar^2/2m) d^2\psi/dx^2$$

Guardrail:

3C does not derive mass m , $\hbar$, physical Hamiltonians, potentials, QFT, or physical 3T existence.

Next recommended chapter:

Toy Model 3D - Adding Potentials as Position-Dependent Phase Rotation Rate.